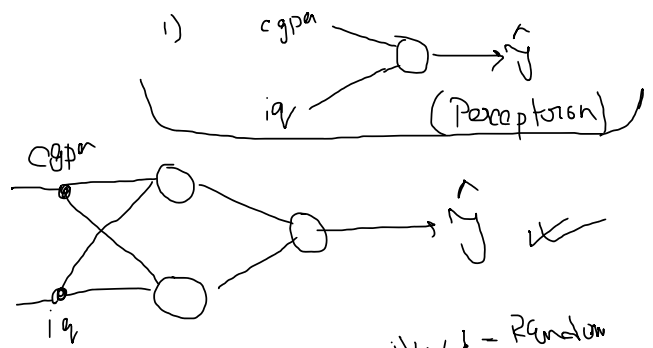


Defⁿ

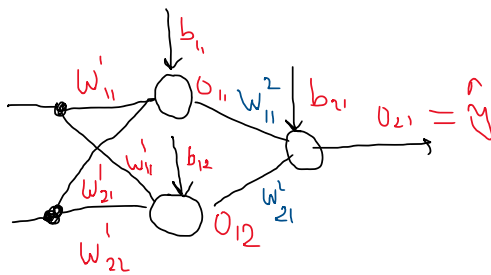
Algorithm → to train Neural Network

x_1 cgpa	x_2 ia	y placement (lpa)
8	80	8
7	70	7

ii)

 $i = \text{row no.}$

$i=2$	$i=1$
7	8
70	80

 x_{i1} x_{i2} 

Steps :-

- 1) w, b - Random
- ii) $w=1, b=0$
- iii)

1. Initialize w, b
2. Select a row (i)
3. Predict using forward Propⁿ.
(Dot Product)

4. Choose a Loss Function.

For this, (MSE)

$$L = (y - \hat{y})^2 = (3 - 18)^2 = 225$$

$$L = f(\hat{y}) \quad \text{const}$$

$$L = f(\hat{y})$$

$$\hat{y} = o_{21} = w_{11}^2 o_{11} + w_{21}^2 o_{12} + b_{21}$$

$$= w_{11}^2 (w_{11}^1 x_{i1} + w_{21}^1 x_{i2} + b_{11}) + w_{21}^2 (w_{12}^1 x_{i1} + w_{22}^1 x_{i2} + b_{12}) + b_{21}$$

$$\hat{y} = f(w, b)$$

5. weight and bias update
→ gradient Decent

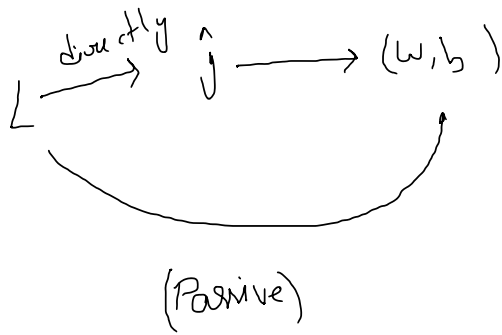
$$w_n = w_0 - \eta \frac{dL}{dw_n}$$

$$L = f(\hat{y}) = f(f(w, b))$$

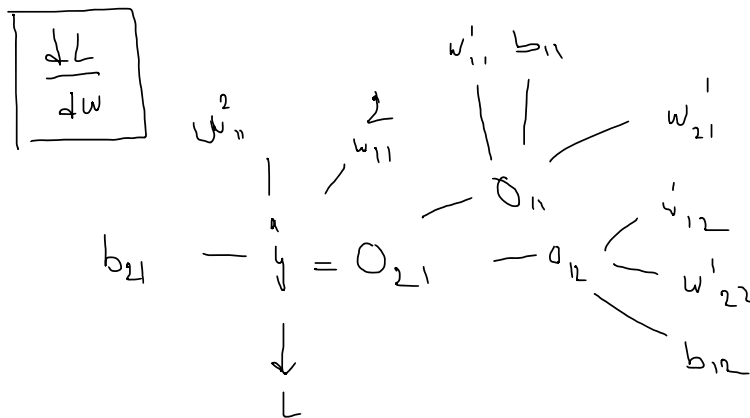
$$L = f(\hat{y}) = f(f(w, b))$$

$$w_n = w_0 - \eta \frac{dL}{dw_n}$$

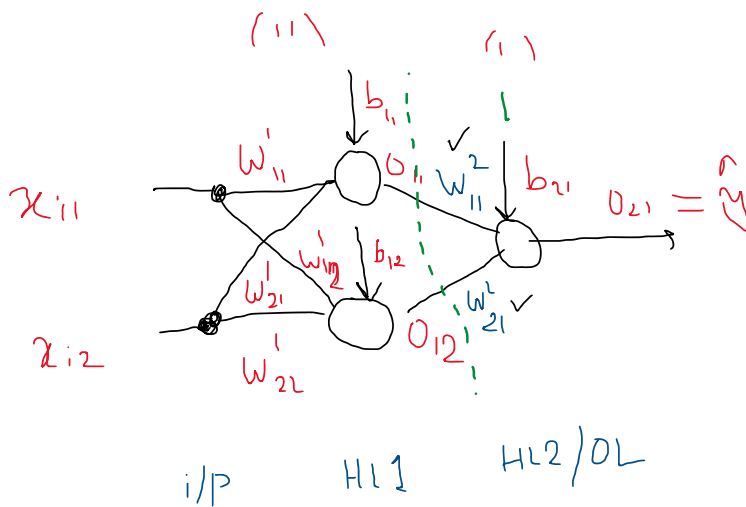
$$b_n = b_0 - \eta \frac{dL}{db_n}$$



$$(w, b) \xrightarrow{\Delta} \hat{y} \xrightarrow{\Delta} L$$



← Hierarchy



$$L \rightarrow \hat{y} \rightarrow w'_{11}$$

chained diff

$$\frac{dL}{dw'_{11}} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dw'_{11}} = -2(y - \hat{y})o_{11}$$

$$\frac{dL}{dw'_{21}} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dw'_{21}} = -2(y - \hat{y})o_{12}$$

$$L = (y - \hat{y})^2$$

$$\frac{dL}{d\hat{y}} = 2(y - \hat{y}) \cdot (-1)$$

$$\frac{dL}{d\hat{y}} = 2(y - \hat{y}) \cdot (-1) \\ = -2(y - \hat{y})$$

$$\hat{y} = w_{11}^2 o_{11} + w_{21}^2 o_{12} + b_{21}$$

$$\frac{d\hat{y}}{dw_{11}^2} = \frac{d}{dw_{11}^2} (w_{11}^2 o_{11} + w_{21}^2 o_{12} + b_{21}) \quad \left| \quad \frac{d\hat{y}}{dw_{21}^2} = o_{12} \right.$$

$$= o_{11}$$

$$\frac{d\hat{y}}{db_{21}} = 1$$

$$\frac{d\hat{y}}{do_{11}} = w_{11}^2$$

$$\hat{y} = w_{11}^2 o_{11} + w_{21}^2 o_{12} + b_{21}$$

$$\hat{y} \Rightarrow w_{11}^2 (w_{11}^1 x_{i1} + w_{21}^1 x_{i2} + b_{11}) + w_{21}^2 (w_{12}^1 x_{i1} + w_{22}^1 x_{i2} + b_{12}) + b_{21}$$

$$\frac{dL}{dw_{11}^2} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{do_{11}} \cdot \frac{do_{11}}{dw_{11}^2} = -2(y - \hat{y}) w_{11}^2 x_{i1}$$

$$L \rightarrow \hat{y} \xrightarrow{o_{11}} w_{11}^1$$

$$\quad \quad \quad \searrow w_{21}^1$$

$$\frac{dL}{dw_{21}^2} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{do_{11}} \cdot \frac{do_{11}}{dw_{21}^2} = -2(y - \hat{y}) w_{11}^2 x_{i2}$$

$$L \rightarrow \hat{y} \xrightarrow{o_{12}} w_{12}^1$$

$$\quad \quad \quad \searrow w_{22}^1$$

$$\hat{y} = w_{11}^2 o_{11} + w_{21}^2 o_{12} + b_{21}$$

$$\hat{y} \Rightarrow w_{11}^2 (w_{11}^1 x_{i1} + w_{21}^1 x_{i2} + b_{11}) + w_{21}^2 (w_{12}^1 x_{i1} + w_{22}^1 x_{i2} + b_{12}) + b_{21}$$

$$L = \frac{1}{2} \sum (y - \hat{y})^2$$

$$\hat{y} \Rightarrow w_{11}^2 (w_{11}' x_{i1} + w_{21}' x_{i2} + b_{11}) + w_{21}^2 (w_{12}' x_{i1} + w_{22}' x_{i2} + b_{12}) + b_{21}$$

$$\frac{dL}{dw_{12}'} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dO_{12}} \cdot \frac{dO_{12}}{dw_{12}'} = -2(y - \hat{y}) \cdot w_{21}^2 x_{i1}$$

$$\frac{dL}{dw_{22}'} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dO_{12}} \cdot \frac{dO_{12}}{dw_{22}'} = -2(y - \hat{y}) \cdot w_{21}^2 x_{i2}$$

$$\frac{dL}{db_{11}} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dO_{11}} \cdot \frac{dO_{11}}{db_{11}} = -2(y - \hat{y}) \cdot w_{11}^2$$

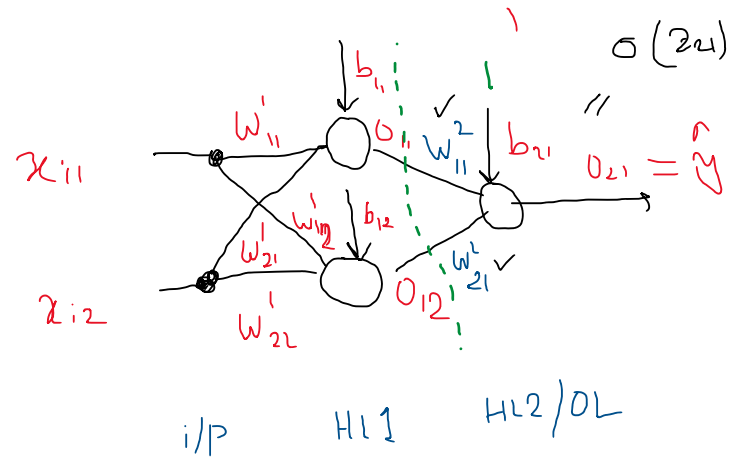
$$\frac{dL}{db_{12}} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dO_{12}} \cdot \frac{dO_{12}}{db_{12}} = -2(y - \hat{y}) \cdot w_{12}^2$$

(Regression)

Classification

18 December 2025 20:27

rgpa	iq	placement
7	80	0
8	70	1
9	92	1
5	70	0



$$L = -y \log \hat{y} - (1-y) \log(1-\hat{y}) \leftarrow \text{Binary CE}$$

$b=0$
 $w=1$

$$z_{11} = 7 \times 1 + 80 \times 1 + 0 = 87$$

$$O_{11} = \sigma(z_{11})$$

$$\frac{dL}{d\hat{y}} = -\frac{y}{\hat{y}} + \frac{(1-y)}{(1-\hat{y})}$$

$$\frac{dL}{d\hat{y}} = -\frac{(y-\hat{y})}{\hat{y}(1-\hat{y})}$$

$$\frac{dL}{dw_{11}^2} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dz_{21}} \cdot \frac{dz_{21}}{dw_{11}^2}$$

$$= -(y-\hat{y}) O_{11} \checkmark$$

$$\frac{dL}{dw_{11}^2} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dz_{21}} \cdot \frac{dz_{21}}{dw_{11}^2}$$

$$L \leftarrow \hat{y} \leftarrow O_{21} \leftarrow w_{11}^2$$

$$L \leftarrow \hat{y} \leftarrow O_{21} \leftarrow z_{21} \leftarrow w_{11}^2$$

$$\hat{y} = O_{21} = \sigma(z_{21})$$

$$\hat{y} = \sigma(z_{21})$$

$$\frac{d\hat{y}}{dz_{21}} = \sigma(z_{21})(1-\sigma(z_{21}))$$

$$z_{21} = \hat{y}(1-\hat{y})$$

$$z_{21} = w_{11}^1 O_{11} + w_{11}^2 O_{12}$$

$$\frac{dL}{dw_{21}} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dz_{21}} \cdot \frac{dz_{21}}{dw_{21}}$$

$$= -(y - \hat{y}) \cdot 0_{12} \quad \checkmark$$

$$\frac{dL}{db_{21}} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dz_{21}} \cdot \frac{dz_{21}}{db_{21}}$$

$$= -(y - \hat{y}) \quad \checkmark$$

$$\frac{dL}{dw'_{11}} = 1 \leftarrow \hat{y} \leftarrow z_{21} \leftarrow 0_{11}$$

$$\leftarrow z_{11} \leftarrow \begin{cases} w'_{11} \\ w'_{21} \\ b_{11} \end{cases}$$

$$\frac{dL}{dw'_{11}} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dz_{21}} \cdot \frac{dz_{21}}{d0_{11}}$$

$$\frac{d0_{11}}{dz_{11}} \cdot \frac{dz_{11}}{dw'_{11}}$$

$$\frac{dL}{dw'_{12}} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dz_{21}} \cdot \frac{dz_{21}}{d0_{12}}$$

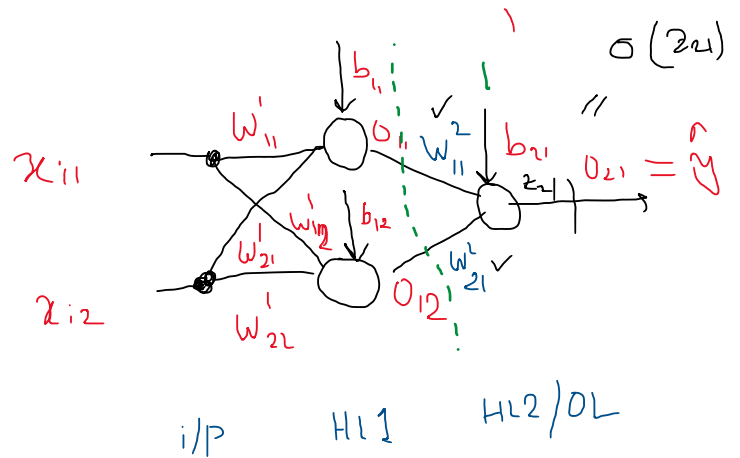
$$\frac{d0_{12}}{dz_{12}} \cdot \frac{dz_{12}}{dw'_{12}}$$

$$z_{21} = w'_{11} \cdot 0_{11} + w'_{21} \cdot 0_{21} + b_{21}$$

$$\frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dz_{21}}$$

$$= - \frac{(y - \hat{y})}{\hat{y}(1 - \hat{y})}$$

$$= -(y - \hat{y})$$



$$z_{12} =$$

$$\frac{dL}{dw'_{12}} \cdot \frac{dL}{dw'_{22}} \cdot \frac{dL}{db_{12}}$$

u

$$\frac{\partial O_{11}}{\partial Z_{11}} \cdot \frac{\partial Z_{11}}{\partial w_{21}}$$

$$\frac{\partial L}{\partial b_{11}} = \frac{\partial L}{\partial y} \cdot \frac{\partial y}{\partial z_{21}} \cdot \frac{\partial z_{21}}{\partial O_{11}}$$

$$\frac{\partial O_{11}}{\partial Z_{11}} \cdot \frac{\partial Z_{11}}{\partial b_{11}}$$

total trainable param:- 9 6 w
3 b

[illegible]

↳ fancy word for derivative.

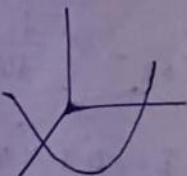
$$\frac{dy}{dx} = \frac{d}{dx}(f(x)) = \frac{d}{dx}(x^2 + 2) = 2x + 1$$

$y \rightarrow x \rightarrow \text{derivative} \quad \frac{d}{dx}$

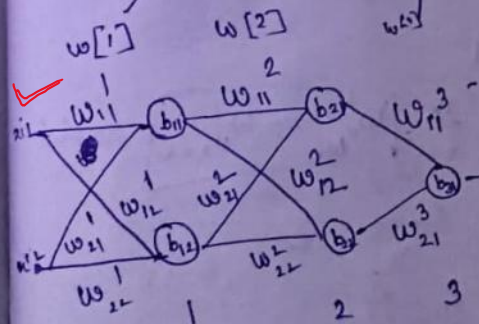
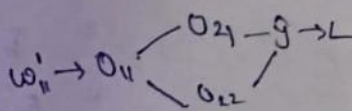
multiple params

$$f(x,y) = x^2 + y^2$$
$$\frac{\partial f}{\partial x} = 2x \quad \frac{\partial f}{\partial y} = 2y$$

gradient $\frac{\partial}{\partial x}$



← 3D.



$$\frac{dL}{d\omega_{11}^3} = \frac{dL}{dy} \cdot \frac{dy}{d\omega_{11}^3}$$

$$\frac{dL}{d\omega_H^L} = \frac{dL}{dy} \cdot \frac{dy}{d\theta_{21}} \cdot \frac{d\theta_{21}}{d\omega_H^L} \quad \checkmark$$

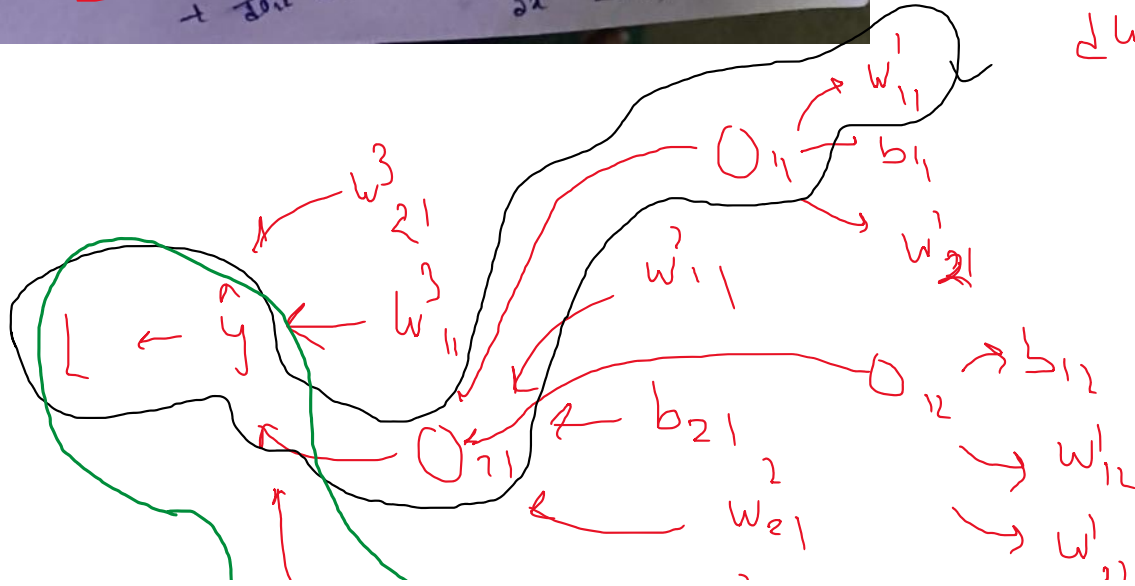
$$L \rightarrow \hat{y} \rightarrow 0_{11} \rightarrow 0_{11} \rightarrow 0_{11}$$

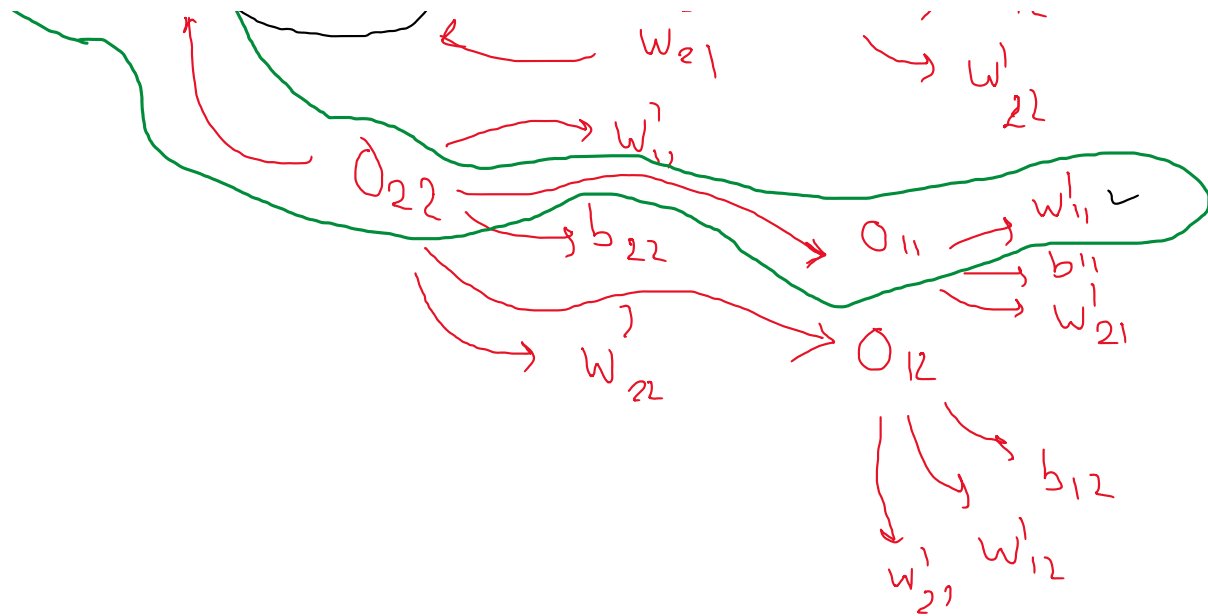
$$\frac{dL}{dw''} = \frac{\partial L}{\partial y} \left[\frac{\partial y}{\partial a_1} \frac{\partial y}{\partial a_2} \frac{\partial y}{\partial w''} \right] \left| \begin{array}{l} \rightarrow f(n) \\ \rightarrow g(n) \end{array} \right. \rightarrow h(p(n), g(n))$$

$$\frac{dL}{dw}^3$$

$$\frac{dL}{dW_{11}}$$

$$\frac{dL}{dW_{11}}$$





$$\frac{dL}{dw'_{11}} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dO_{21}} \cdot \frac{dO_{21}}{dO_{11}} \cdot \frac{dO_{11}}{dw'_{11}} + \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dO_{11}} \cdot \frac{dO_{21}}{O_{11}} \cdot \frac{dO_{11}}{dw'_{11}}$$

backpropagation = chained derivation
 $\Rightarrow$ memoization